AI Ethics and Regulation

BMED365: Topic List A01–A09

BMED365

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Overview

A01: Discuss the four bioethical principles in an AI context

Beauchamp & Childress' Four Principles:

1. **Autonomy** (Respect for self-determination)
   - Informed consent for AI use
   - Right to know if AI is involved
   - Ability to opt out of AI decisions
2. **Beneficence** (Do good)
   - AI should improve patient outcomes
   - Validated and effective
   - Accessible to all
3. **Non-maleficence** (Primum non nocere)
   - Avoid harm from incorrect predictions
   - Minimize bias and discrimination
   - Safety testing before clinical use
4. **Justice** (Fair distribution)
   - Equal access to AI-enhanced healthcare
   - Avoid systematic discrimination
   - Representative training data

Challenge

The principles can conflict – e.g. efficiency (beneficence) vs. privacy (autonomy)

A02: Identify types of bias in medical AI systems

Bias = systematic skew leading to unfair outcomes

Sources of bias:

1. **Historical bias:** Training data reflects historical unfairness
   - Example: Underrepresentation of minorities in clinical studies
2. **Representation bias:** Skewed patient population in training data
   - Example: Model trained only on white patients
3. **Measurement bias:** Incorrect proxy variables
   - Example: Using healthcare costs as a proxy for disease severity (correlates with race)
4. **Evaluation bias:** Test data is not representative
5. **Aggregation bias:** One model for all, ignoring subgroups

Consequence

Bias can cause AI systems to provide worse healthcare to already marginalized groups.

A03: Explain privacy considerations when using LLMs

Privacy challenges with LLMs:

Data leakage to third parties:

- Input is sent to LLM provider
- Potentially used for retraining
- Stored on foreign servers

Memorization:

- LLMs may have "learned" personal data
- Can generate sensitive info in output

Identification:

- Re-identification from seemingly anonymous data
- Prompts can reveal context

Countermeasures:

- Never share patient-identifiable info with public LLMs
- Use local/secure solutions (e.g. chat.uib.no)
- Anonymization and pseudonymization

Important rule

Never enter patient data into ChatGPT, Claude or other public LLMs!

A04: Describe the main features of the EU AI Act

EU AI Act (adopted 2024):

- World's first comprehensive AI regulation
- Risk-based approach
- Applies to all who offer or use AI in the EU

Main principles:

1. **Risk classification:** AI systems categorized by risk level
2. **Requirements proportional to risk:** Higher risk = stricter requirements
3. **Transparency:** Users must be informed about AI use
4. **Human oversight:** Human-in-the-loop for high-risk systems
5. **Documentation:** Technical documentation and logging
6. **CE marking:** Conformity assessment before deployment

Timeline

Gradual implementation 2024–2027. High-risk AI requirements apply from 2026.

A05: Explain risk classification in the EU AI Act

Four risk levels:

1. **Unacceptable risk** (PROHIBITED)
   - Social scoring, manipulation, real-time facial recognition
2. **High risk** (STRICT REQUIREMENTS)
   - Medical devices and diagnostics
   - Critical infrastructure, education, labor market
   - Requires risk assessment, documentation, human oversight
3. **Limited risk** (TRANSPARENCY REQUIREMENTS)
   - Chatbots, deepfakes
   - Must inform users that it is AI
4. **Minimal risk** (NO SPECIFIC REQUIREMENTS)
   - Spam filters, game AI
   - Voluntary ethical guidelines

Medical AI

Most medical AI systems will be classified as **high risk**.

A06: Discuss requirements for high-risk AI in healthcare

Requirements for high-risk AI systems:

Before deployment:

- Risk assessment and management
- High data quality and representativeness
- Technical documentation
- Transparency and user guidance
- Cybersecurity
- Conformity assessment (CE marking)

During use:

- Automatic logging
- Human oversight
- Post-market surveillance
- Incident reporting
- Periodic re-evaluation

Specific to medical AI:

- Coordinated with MDR (Medical Devices Regulation)
- Clinical evidence and validation required
- Requirements for explainability and robustness

A07: Reflect on responsibility allocation when AI fails

Who is responsible when AI makes mistakes?

Potential responsible parties:

- **AI developer:** Design, training, validation
- **Healthcare institution:** Procurement, implementation, training
- **Clinician:** Uses AI, makes final decision
- **Patient:** Consents to AI use?

Key issues:

- AI as "black box" – difficult to understand errors
- Shared responsibility $\rightarrow$ diluted responsibility?
- Automation bias – over-reliance on AI

Legal developments:

- AI Liability Directive (EU)
- Product liability extended to AI
- Burden of proof may be placed on AI provider

Core principle

The clinician retains ultimate responsibility – AI is a **tool**, not a replacement.

A08: Know about GDPR-relevant aspects of AI

GDPR and AI in healthcare:

Key principles:

- **Lawful basis:** Consent, necessity, legitimate interest
- **Purpose limitation:** Data only for stated purpose
- **Data minimization:** Only necessary data
- **Storage limitation:** Do not store longer than necessary

Special provisions:

- **Art. 22:** Right not to be subject to automated decisions
- **Art. 9:** Health data is a "special category" – extra protected
- **Art. 35:** DPIA (Data Protection Impact Assessment) for high-risk processing

Important

Patient data used for AI training/use requires a lawful basis and often a DPIA.

A09: Discuss algorithmic fairness

Fairness in ML: No universal definition

Different fairness definitions:

1. **Demographic parity:** Equal proportion of positive predictions per group
2. **Equalized odds:** Equal TPR and FPR per group
3. **Calibration:** Confidence scores mean the same for all groups
4. **Individual fairness:** Similar individuals are treated similarly

The problem:

- These definitions are often **mathematically incompatible**
- Choice of definition is a **value-laden** decision
- "Fair" in one sense can be "unfair" in another

Medical example

Should a risk model give the same score to a 30-year-old and a 70-year-old with the same symptoms? Age is a risk factor – is that discrimination?

Summary: AI Ethics and Regulation

Bioethics and bias:

- A01: Four bioethical principles in an AI context
- A02: Types of bias – historical, representation, measurement, evaluation
- A03: Privacy considerations with LLMs

EU AI Act:

- A04–A06: Risk-based approach, high-risk requirements for medicine

Responsibility and Fairness:

- A07: Responsibility allocation – complex, clinician has ultimate responsibility
- A08: GDPR – special protection for health data
- A09: Algorithmic fairness – no simple solution

Practical work in Lab 3

Discuss ethical implications of LLMs, bias analysis, and privacy considerations.